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United States Patent

12403338

Kind Code

B2

Date of Patent

September 02, 2025

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Filtering facepiece respirator with an adjustable inhalation-exhalation valve

Abstract

A filtering facepiece respirator is provided. The filtering facepiece respirator includes a facemask adapted to fit over the nose and mouth of a wearer, where the facemask comprises a mask body containing a filtering structure. A harness is coupled to the mask body for securing the facemask on the face of the wearer, and a mechanical valve is coupled to a portion of the mask body proximate the wearer's mouth. The valve may be adjusted between a first mode of operation and a second mode of operation, where in the first mode of operation is permitted to flow through the valve on exhalation but not on inhalation and in the second mode of operation air is obstructed from flowing through the valve on exhalation and inhalation. There is an optional third mode of operation where air is permitted to flow through the valve on inhalation and exhalation.

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Family ID: 80739749

Appl. No.: 17/462351

Filed: August 31, 2021

Prior Publication Data

Document Identifier

US 20220088423 A1

Publication Date

Mar. 24, 2022

Related U.S. Application Data

us-provisional-application US 63082443 20200923

Publication Classification

Int. Cl.: **A62B18/10** (20060101); **A41D13/11** (20060101); **A61M16/20** (20060101); **A62B9/00** (20060101); **A62B18/08** (20060101); **A62B23/02** (20060101); **F16K15/14** (20060101); **F16K15/18** (20060101); **H04Q9/00** (20060101)

U.S. Cl.:

CPC **A62B18/10** (20130101); **A41D13/11** (20130101); **A61M16/202** (20140204); **A62B9/006** (20130101); **A62B18/084** (20130101); **A62B23/025** (20130101); **F16K15/148** (20130101); **F16K15/1825** (20210801); **H04Q9/00** (20130101); **H04Q2209/40** (20130101)

Field of Classification Search

CPC: A62B (18/10); A62B (23/025); A62B (18/086); A62B (27/00); F16K (15/148); F16K (15/1825); A63B (23/18)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority under 35 USC § 119(e) to U.S. Provisional Patent Application No. 63/082,443, filed on Sep. 23, 2020, titled FILTERING FACEPIECE RESPIRATOR WITH AN ADJUSTABLE INHALATION-EXHALATION VALVE, the disclosure of which is hereby incorporated by reference in its entirety for all purposes.

TECHNICAL FIELD

(1) The present disclosure generally relates to personal protective equipment, and more specifically, to a filtering facepiece respirator that uses an adjustable inhalation-exhalation valve that is integrated into the mask body.

BACKGROUND

(2) Filtering facepiece respirators are commonly worn over the mouth and nose of a person to prevent the individual from inhaling and/or exhaling microscopic particles, including infectious and non-infectious contaminants. The use of these devices has become much more common due to the recent SARS-CoV-2 (COVID-19) pandemic.

(3) Conventional filtering facepiece respirators have either no valve or a valve that filters on inhalation but not on exhalation. These masks typically use a fine mesh of nonwoven polypropylene fabric for filtration. Wearing a non-valved respirator can trap exhaled air with a lower oxygen concentration and a higher carbon dioxide level than room air. This trapped air is then repeatedly rebreathed, causing hypoxemia and hypercapnia. The resulting acute symptoms of these alterations in oxygen and carbon dioxide include dizziness and headaches, causing a reduction in work efficiency and can negatively affect decision making. Chronic hypoxemia and hypercapnia can exacerbate pre-existing cardiac, pulmonary, vascular, neurologic, and metabolic conditions.

(4) Prior to the COVID-19 pandemic, over 3 million United States employees in approximately 1.3 million workplaces were required to wear some kind of respiratory protection. Since early 2020, this number has increased greatly. An N95 filtering facepiece respirator (FFR) is a type of respirator which removes particles from the air that are breathed through it. These respirators filter out at least 95% of very small (0.3 microns) particles. Other respirator filter classes include N99, N100, R95, R99, R100, P95, P99, and P100. N95 FFRs are capable of filtering out all types of particles, including bacteria and viruses. The vast majority of these respirators do not have an expiratory valve. An N95 respirator with an exhalation valve provides a similar level of protection to the wearer as one that does not have a valve. The presence of an exhalation valve reduces exhalation resistance, which makes it easier to breathe on exhalation. A respirator with an exhalation valve keeps the face cooler, reduces moisture build-up inside the facepiece, and does not trap exhaled air as readily as a non-vented respirator. However, respirators with exhalation valves should not be used in many situations. Examples include but are not limited to times where a sterile field must be maintained (e.g., during an invasive procedure in a surgical suite) or when in close proximity to another individual when sick or during a pandemic (e.g., when working in a meatpacking plant or visiting a hospital during a pandemic or when sick) because the exhalation valve may allow unfiltered contaminated exhaled air to escape. If one only has a respirator with an exhalation valve but requires filtration on exhalation, the valve can be covered with another mask (surgical or procedure mask) that does not interfere with the respirator fit. This approach, however, wastes personal protective equipment. The mask design described in this invention would not require the placement of a second mask. The user would change the setting on the switch or have it changed remotely to close the exhalation valve.

(5) The Occupational Safety and Health Administration (OSHA) requires an annual fit test to confirm the fit of any respirator that forms a tight seal on the wearer's face before it is used in the workplace. Once a fit test has been done to determine the best respirator model and size for a particular user, a user seal check should be done every time the respirator is to be worn to ensure an adequate seal is achieved. Presently, medical and nonmedical users of these non-vented masks have to physically remove them to inhale fresh air. A major issue arises in medical settings when healthcare providers remove a mask that is contaminated. It is not uncommon for these individuals to remove their masks numerous times per day, and this can lead to the transmission of a pathogen to the mask user or another individual. Each and every time the respirator is removed, another user

seal check must be performed. Adding an adjustable valve that will allow the free flow of air out or in and out of the mask will circumvent the need for removal to clear trapped air or get a breath of fresh air and reverse hypoxia and hypercapnia. This increases safety by decreasing the chance of transferring infectious pathogens and adds to work efficiency by omitting repeated user seal checks.

(6) A need therefore exists for a filtering facepiece respirator that addresses the aforesaid challenges yet is easy to operate in various modes of operations depending on the wearer's desired need.

SUMMARY OF THE INVENTION

(7) A manually operable filtering facepiece respirator is described herein. The filtering facepiece respirator includes a facemask adapted to fit over the nose and mouth of a wearer, where the facemask comprises a mask body containing a filtering structure. A harness is coupled to the mask body for securing the facemask on the face of the wearer, and a mechanical valve is coupled to a portion of the mask body proximate the wearer's mouth. The valve may be manually adjusted between a first mode of operation and a second mode of operation, where in the first mode of operation air is permitted to flow through the valve on exhalation but not on inhalation (vented mode), and in the second mode of operation air is obstructed from flowing through the valve on exhalation and inhalation (closed mode). In some embodiments, the valve can operate in a third mode where air is permitted to flow through the valve when the wearer inhales or exhales (open mode).

(8) In some embodiments, the mask body comprises a first porous layer and a filtering layer. In other embodiments, the mask body comprises a first porous layer, a second porous layer, and a filtering layer, where the filtering layer is disposed between the first porous layer and the second porous layer.

(9) The mechanical valve comprises a valve housing having an interior chamber, a fixed hub member disposed within the chamber, a first movable hub member disposed within the chamber, a pliable diaphragm disposed between the fixed hub member and the first movable hub member, a screw member disposed within the chamber extending along a longitudinal axis of the valve housing, and an adjustable dial. The adjustable dial is rotatable about the longitudinal axis, where the adjustable dial is coupled to the screw member and the screw member is threadedly coupled to the first movable hub member such that when the adjustable dial is rotated by the wearer, the screw member is also rotated and the threaded engagement between the screw member and the first movable hub member causes the first movable hub member to translate axially along the longitudinal axis away from or towards the fixed hub member.

(10) In the first mode of operation, the first movable hub member is spaced apart from the fixed hub member, thus permitting an outer periphery of the diaphragm to be urged away from the fixed hub member under positive pressure, and air is allowed to freely flow through the valve when the wearer exhales. However, the outer periphery of the diaphragm is drawn towards the fixed hub member under negative pressure to obstruct the flow of air through the valve when the wearer inhales.

(11) In the second mode of operation, the first movable hub member abuts the fixed hub member to secure the diaphragm therebetween. This retains an outer periphery of the diaphragm in contact with the fixed hub member to create a seal therebetween that obstructs air from flowing through the valve when the wearer exhales or inhales.

(12) In some embodiments, the valve may operate in a third mode of operation. In this mode, air is permitted to flow through the valve when the wearer inhales or exhales.

(13) The valve may be adjusted from one mode of operation to another mode of operation by a switch, slide, rod, rotating arm, lever, button, toggle, dial, knob, joystick, rheostat, or electronic controller.

(14) The filtering facepiece respirator may further include a valve cap indicator coupled to a front

portion of the valve housing. The valve cap indicator comprises a plate of material inscribed with indicia indicating the state of airflow through the valve utilizing a colored indicator, label, light indicator, or display. In some embodiments, the valve cap indicator may include an auditory system, such as a transistor, to broadcast the mode of operation to the wearer.

(15) An automated filtering facepiece respirator is further described herein. The filtering facepiece respirator includes a facemask adapted to fit over the nose and mouth of a wearer, where the facemask comprises a mask body containing a filtering structure. A harness is coupled to the mask body for securing the facemask on the face of the wearer. A mechanical valve is coupled to a portion of the mask body proximate the wearer's mouth. The valve comprises a valve housing having a hollowed interior, a battery coupled to the housing, and a controller disposed within the hollowed interior. The controller is electrically coupled to the battery, where the controller adjusts the valve between a first mode of operation and a second mode of operation, where in the first mode of operation air is permitted to flow through the valve on exhalation but not on inhalation (vented) and in the second mode of operation air is restricted from flowing through the valve on inhalation and exhalation (closed).

(16) In some embodiments, the mask body comprises a first porous layer and a filtering layer. In other embodiments, the mask body comprises a first porous layer, a second porous layer, and a filtering layer, where the filtering layer is disposed between the first porous layer and the second porous layer.

(17) In this example, the valve comprises a valve housing having an interior chamber, a fixed hub member disposed within the chamber, a first movable hub member disposed within the chamber, a pliable diaphragm disposed between the fixed hub member and the first movable hub member, and a screw member disposed in the chamber extending along a longitudinal axis of the valve housing. The controller is in electrical communication with a motor coupled to the screw member to rotate the screw member. The screw member is threadedly coupled to the first movable hub member such that when the screw member is rotated by the motor, the threaded engagement between the screw member and the first movable hub member causes the first movable hub member to translate axially along the longitudinal axis away from or towards the fixed hub member.

(18) In the first mode of operation, the first movable hub member is spaced apart from the fixed hub member, thus permitting an outer periphery of the diaphragm to be urged away from the fixed hub member under positive pressure and air is allowed to flow through the valve when the wearer exhales. But the outer periphery of the diaphragm is drawn towards the fixed hub member under negative pressure to obstruct the flow of air through the valve when the wearer inhales.

(19) In the second mode of operation the, first movable hub member abuts the fixed hub member to secure the diaphragm therebetween, thus retaining an outer periphery of the diaphragm in contact with the fixed hub member to create a seal therebetween that obstruct air from flowing through the valve when the wearer exhales or inhales.

(20) In some embodiments, the controller may be controlled by a mobile device via Bluetooth, WiFi, cellular, ultra-wideband, or RFID communications. In some embodiments, the controller may be integrated for use within public spaces such as offices, hospitals, clinics, restaurants, public and private buildings, service businesses, gyms and health clubs, and retailers.

(21) In some embodiments, the filtering facepiece respirator may also include smart sensors that detect oxygen saturation levels, carbon dioxide levels, and volume of expired or inspired air. In these embodiments, the smart sensors may report data to a central monitoring unit, where the central monitoring unit uses the smart sensor data to alert the user of a change in mask environment conditions.

(22) Other devices, apparatus, systems, methods, features, and advantages of the disclosure will be or will become apparent to one with skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, and be protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The present invention may be better understood by referring to the following figures.

Implementations of the invention are illustrated by way of example only and not limitation in the figures of the accompanying drawings. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the disclosure. In the figures, like reference numerals designate corresponding parts throughout the different views.

(2) FIG. 1 is a perspective view of an example of a manual tri-mode filtering facepiece respirator in accordance with the teachings of the present disclosure.

(3) FIG. 2 is a cross-sectional view of the tri-mode filtering facepiece respirator of FIG. 1.

(4) FIG. 3 is an exploded view of a respiratory valve of the tri-mode filtering facepiece respirator of FIG. 1.

(5) FIG. 4A is a front view of the respiratory valve of FIG. 3 in an “open” mode of operation.

(6) FIG. 4B is a cross-sectional view of the respiratory valve of FIG. 3 in an “open” mode of operation.

(7) FIG. 4C is a partial close-up view of the elongated slot feature of the respiratory valve of FIG. 3.

(8) FIG. 5A is a front view of the respiratory valve of FIG. 3 in a “vented” mode of operation.

(9) FIG. 5B is a cross-sectional view of the respiratory valve of FIG. 3 in a “vented” mode of operation.

(10) FIG. 6A is a front view of the respiratory valve of FIG. 3 in a “closed” mode of operation.

(11) FIG. 6B is a cross-sectional view of the respiratory valve of FIG. 3 in a “closed” mode of operation.

(12) FIG. 7 is a perspective view of an example of a motorized bi-mode filtering facepiece respirator in accordance with the teachings of the present disclosure.

(13) FIG. 8 is an exploded view of a respiratory valve of the bi-mode filtering facepiece respirator of FIG. 7.

(14) FIG. 9A is a front view of the respiratory valve of FIG. 8 in a “vented” mode of operation.

(15) FIG. 9B is a cross-sectional view of the respiratory valve of FIG. 8 in a “vented” mode of operation.

(16) FIG. 10A is a front view of the respiratory valve of FIG. 8 in a “closed” mode of operation.

(17) FIG. 10B is a cross-sectional view of the respiratory valve of FIG. 8 in a “closed” mode of operation.

(18) FIG. 11 is a perspective view of an example of a manual bi-mode facepiece respirator with a rotating valve controller, in accordance with the teachings of the present disclosure.

(19) FIG. 12A is a front view of a respiratory valve of the bi-mode filtering facepiece respirator of FIG. 11 in a “vented” mode of operation.

(20) FIG. 12B is a cross-sectional view of the respiratory valve of FIG. 12A in a “vented” mode of operation.

(21) FIG. 13A is a front view of the respiratory valve of FIG. 12A in a “closed” mode of operation.

(22) FIG. 13B is a cross-sectional view of the respiratory valve of FIG. 12A in a “closed” mode of operation.

(23) FIG. 14 is a perspective view of an example of a manual bi-mode sliding controller valve facepiece respirator in accordance with the teachings of the present disclosure.

(24) FIG. 15 is an exploded view of a respiratory valve of the bi-mode filtering facepiece respirator of FIG. 14.

(25) FIG. 16A is a front view of the respiratory valve of FIG. 15 in a “vented” mode of operation.

(26) FIG. 16B is a cross-sectional view of the respiratory valve of FIG. 15 in a “vented” mode of operation.

operation.

(27) FIG. 17A is a front view of the respiratory valve of FIG. 15 in a “closed” mode of operation.

(28) FIG. 17B is a cross-sectional view of the respiratory valve of FIG. 15 in a “closed” mode of operation.

DETAILED DESCRIPTION

(29) FIGS. 1-17B illustrate examples of various embodiments of a filtering facepiece respirator. Generally, the filtering facepiece respirator includes a facemask adapted to fit over the nose and mouth of a wearer, where the facemask comprises a mask body containing a filtering structure. A harness is coupled to the mask body for securing the facemask on the face of the wearer, and a mechanical valve is coupled to a portion of the mask body proximate the wearer's mouth. The valve may be manually adjusted between a first mode of operation and a second mode of operation, where in the first mode of operation air is permitted to flow through the valve on exhalation but not on inhalation (vented mode) and in the second mode of operation air is obstructed from flowing through the valve on inhalation and exhalation (closed mode). In some implementations, the valve may be further adjusted to a third mode of operation, where in the third mode of operation air is permitted to flow through the valve when the wearer inhales and exhales (open mode)

(30) FIG. 1 is a perspective view of one example embodiment of a manual tri-mode filtering facepiece respirator **100** according to the teaching of the present disclosure. As shown, the facepiece respirator **100** may include a facemask **102**, an upper harness **104** coupled to an upper portion of the face mask **102**, a lower harness **106** coupled to a lower portion of the facemask **102**, and a respiratory valve **108** coupled to a central front portion of the facemask **102**. In preferred implementations, the facemask **102** may include an outer contour **110** defined about a peripheral edge of the facemask **102** to substantially cover the nose and mouth of a wearer.

(31) In most embodiments, the facemask **102** may further include a bendable reinforcement nosepiece **112** coupled to an upper portion of the outer contour **110**. The nosepiece **112** may be coupled to the outer contour **110** of the facemask **102**, for example, by glue, bonding, or other means at a portion of the facemask **102** fitting over the nose of the wearer. The nosepiece **112** may be constructed from a strip of aluminum, plastic, or any other pliable material. The nosepiece **112** is preferably flexible to allow the user to adjust the fit of the facemask **102** around the nose bridge area of the wearer. While the nosepiece **112** is shown affixed to the outside of the facemask **102**, in other embodiments, the nosepiece **112** may be affixed to the outer contour **110** along the inside of the facemask **102** or embedded within the facemask **102** structure.

(32) FIG. 2 is a cross-section view of filtering facepiece respirator **100**. As shown, the facemask **102** may include a substantially dome-shaped mask body **202** comprising an outer porous layer **204**, an inner porous layer **206**, a filtering layer **208**, and a valve port **210** formed in a front central portion of the mask body **202**. The inner layer **206** defines an interior cavity **212** for enclosing the nose and mouth of the wearer.

(33) The outer porous layer **204** may be molded using polyester fibers. The layer serves as an outer support layer and pre-filter. The inner porous layer **206** may also be formed as a fiberfill shell and may also be molded using polyester fibers. While the outer porous layer **204** and the inner porous layer **206** are described herein as being made from polyester fibers, in other implementations, the outer porous layer **204** and the inner porous layer **206** may be made from any other permeable fabric or material.

(34) The inner and outer layers **204** and **206** may be made of different or the same material. For example, the outer layer **204** may be formed of larger fibers than the inner layer **206** so that the inner layer **206**, by being formed of finer fibers, may have a softer and therefore more comfortable surface to lie against the face of the wearer.

(35) As shown, the inner and outer layers **204** and **206** are configured to sandwich the filtering layer **208**. The filtering layer **208** may be formed from any known type of filter material so as to provide for the filtering of particular elements in the air. For example, the filtering layer **208** may

be composed of nonwoven, interlaced polypropylene fibers, but can also be made of electrospun nanofibers, activated charcoal-treated sheets or sheets formed from charcoal particles, fiberglass material, cellulose, or other natural materials. The filtering layer **208** should preferably be electrostatic, hydrophobic, and water and droplet-proof.

(36) In preferred embodiments, the filtering layer **208** should be adapted to trap particles having a size range of known contaminants or pathogens. The filter layer **208** material may be chosen in response to the specific contaminants or pathogens sought to be sieved.

(37) In other embodiments, the filtering layer **208** may comprise two or more layers of material having different filtering specifications, such that one layer filters larger particles and the other layers filter the specific smaller particles of interest. This arrangement may affect various performance parameters of the facemask **102**.

(38) In this way, inner and outer layers **204** and **206** are intended to be more porous than the filtering layer **208** and are constructed to continually sanitize contaminant particles trapped by filtering layer **208**. In addition, the inner and outer layers **204** and **206** may further serve to sanitize contaminants passing therethrough prior to reaching filtering layers **208**.

(39) The use of three layers **204**, **206**, and **208**, as shown in FIG. 2, is illustrative only and may vary depending upon the specific application of the filtering facepiece **100** and its performance specifications. For example, it may be useful to construct a mask using only a single filtering layer **208** and only a single porous layer **204**, such that the filtering layer **208** is located closer to the wearer than layer **204**. Such an arrangement would provide a simple respirator mask that filters particulate contaminants and sanitizes them to protect the wearer of the facepiece. Likewise, a simplified mask may be constructed just using filtering layer **208** and porous layer **206**, such that layer **206** is located closer to the wearer of such a facepiece. This arrangement would provide a surgical-type mask that traps and sanitizes contaminants exhaled by the mask wearer. It should also be kept in mind that layers **204** and **206** may be used in combination with one or more filtering layers **208** to provide a facepiece suitable for serving either as a respirator or a surgical mask. Finally, to aid in facial recognition, in some embodiments, portions of the three layers **204**, **206**, and **208** as shown in FIG. 2 can be replaced with clear or translucent plastic or silicone.

(40) Turning back to FIG. 1, in some embodiments, the outer contour **110** may be lined with a deformable edge member **114** extending around the peripheral edge of the mask body **102**. The edge member **114** may be made of silicone, thermoplastic polyurethane (TPU), polyvinyl, or other suitable material. More specifically, the edge member **114** may be formed from any compressible resilient polymer with either fast or slow recovery properties. The edge member **114** may be coupled to the outer contour **110** by glue, bonding, or other suitable means.

(41) As better shown in FIG. 2, the three layers **204**, **206**, and **208** may be sealed together by any suitable means, such as ultrasonic welding, about the periphery of the mask body **202** and the edge member **114** may be disposed around the sealed periphery of the mask body **202**. The edge member **114** provides a seal between the facemask **102** and the face of the wearer to prevent air from seeping into or out of the interior cavity **212** along the outer contour **110**.

(42) Returning to FIG. 1, the facemask **102** may be secured on the wearer's face by the upper and lower harnesses **104** and **106**. The upper harness **104** may comprise an elastic member **120** made of a band or string of material coupled to an upper portion of the facemask **102** at each of its opposing ends **122**. The elastic member **120** is configured to extend around the back of the head of the wearer. The elastic member **120** may be made of nylon, rubber, cloth, or any other elastic material. The elastic member **120** may be coupled to the facemask **102** by bonding, welding, or other mechanical means.

(43) The lower harness **106** may comprise an elastic member **130** made of a band or string of material coupled to a lower portion of the facemask **102** at each of its opposing ends **132**. The elastic member **130** is configured to extend around the back of the upper neck of the wearer. The elastic member **130** may be made of nylon, rubber, cloth, or any other elastic material. The elastic

member **130** may be coupled to the facemask **102** by bonding, welding, or other mechanical means. (44) The upper and lower harnesses **104** and **106** may be made of elastic material to provide an adjustable fit. In some embodiments, the upper and lower harnesses **104** and **106** may be adjustable about the head of the wearer by adjustable buckle fasteners, fasteners straps, or any other suitable means. In other embodiments, one end of the harness may be coupled to an upper portion of the facemask **102**, while an opposing end of the harness may be coupled to an upper portion of the facemask **102** such that each harness fits around the wearer's ears to secure the facemask on the wearer's face.

(45) As shown in FIGS. **1** and **2**, the respiratory valve **108** may be coupled to a front central portion of the facemask **102**. In this way, while in use, the respiratory valve **108** is positioned in front of the mouth of the wearer. Alternatively, the respiratory valve **108** may be coupled to the side of the facemask to aid in facial recognition when portions of the three layers **204**, **206**, and **208** are replaced with clear or translucent plastic or silicone.

(46) As better shown in FIG. **2**, the respiratory valve **108** is configured to sit within the valve port **210** and the respiratory valve **108** may be fixed to the mask body **202** by bonding, weldment, glue, or any other suitable means. In other embodiments, the respiratory valve **108** may be detachably coupled to the mask body **202** at the valve port **210**. The respiratory valve **108** can therefore be part of a disposable mask, fully reusable mask, or a reusable mask with disposable components.

(47) As shown, valve **108** may have an annular-shaped construction. In other embodiments, valve **108** may have a polygonal-shaped construction. In any embodiment, the shape and dimensions of valve **108** preferable correspond to the shape and dimensions of the valve port **210**.

(48) FIG. **3** is an exploded view of the manual tri-mode respiratory valve **108**. Moving from right to left, the respiratory valve **108** includes an adjustable dial **302**, a valve cap indicator **310**, an outer hub **320**, a washer **330**, a diaphragm **340**, a valve housing **350**, and an inner hub **380**. Each respiratory valve **108** component may be made of plastic, lightweight metal, ceramic, or any other suitable material.

(49) As shown, the adjustable dial **302** may include an arrow-shaped pointer **306** integrally formed with a dial knob **304**. The dial knob **304** should be constructed to suitable dimensions enabling the wearer to turn the dial knob **304** with the wearer's fingers. In addition, the pointer **306** may be configured to complement indicia inscribed on valve cap indicator **310** to indicate to the wearer what operational mode the respiratory valve **108** is operating in. The adjustable dial **302** and its components may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(50) The valve cap indicator **310** may comprise a thin-disc-shaped plate **312** having indicia **314** inscribed on its upper surface **316**. In the embodiment shown, the indicia **314** includes the wording "OPEN," "VENTED," and "CLOSED" to correspond to the three modes of operation of the respiratory valve **108**. In particular, the indicia **314**, when the integrated pointer **306** of the adjustable dial **302** is positioned over the indicia, indicates to the wearer the mode of operation that the valve **108** is currently operating in.

(51) The valve cap indicator **310** further includes a screw hole **318** for passing a screw member therethrough, as further described below, and an outer periphery **319**. The outer periphery **319** is preferably constructed to diametrical dimensions that correspond to the diametrical dimensions of a first open end of the valve housing **350** such that the outer periphery **319** mates with an inner wall of the valve housing **350** to enclose the first end.

(52) The valve cap indicator **310** may be constructed from plastic, aluminum, stainless steel, or any other suitable material. The indicia **314** may be inscribed or etched into the upper surface **316** of the valve cap indicator **310**. In other embodiments, the indicia **314** may include stickers, light-emitting diodes, or may be painted on the upper surface **316** of the valve cap indicator **310**. In some embodiments, the valve cap indicator **310** may include an auditory system, such as a transistor, to broadcast the mode of operation to the wearer.

(53) Moving further downstream, the first movable hub member, outer hub **320** comprises an annular construction having an annual rim **322** coupled to a central hub portion **324** by a series of spokes **326** extending therebetween. The central hub portion **324** includes a threaded screw hole **328** for engaging a screw member extending therethrough, as described in further detail below. Each spoke **326** may be substantially triangular in shape, such that its thickness gradually increases as the spoke **326** extends from the central hub portion **324** towards the annual rim **322**.

(54) The outer hub **320** may further include a series of stabilizing tabs **329** formed about the periphery of the annual rim **322**. The stabilizing tabs **329** are constructed to the dimensions and configured to fit within a corresponding series of elongated slots formed in the annular wall of the valve housing **350** to prevent the outer hub **320** from rotating as it is axially translated within the valve housing **350**.

(55) The outer hub **320** may be constructed to have a relatively thin thickness. For example, outer hub **320** may be constructed to a thickness of approximately 0.125 inches. The outer hub **320** may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(56) The washer **330** may include a thin disc-shaped body **332** having an orifice **334** extending therethrough. The washer **330** secures the diaphragm **340** to the center of the valve housing **350** and serves as a bearing surface between the outer hub **320** and the diaphragm **340**. The washer **330** may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(57) Next, the diaphragm **340** may include a thin disc-shaped body **342** having an orifice **344** extending therethrough. The diaphragm **340**, when secured in close proximity to a central hub housed within the valve housing **350**, forms a hermetic seal to prevent airflow through the respiratory valve **108**, as described in further detail below. The diaphragm **340** may be constructed from rubber, latex, polymers or any other suitable pliable material.

(58) In other embodiments, the diaphragm **340** may be constructed from filtering media. In particular, the diaphragm **340** may be made of porous material that filters microscopic particulates. In such embodiments, air may be being filtered through the facemask **102** and the respiratory valve **108**.

(59) Moving further downstream, the valve housing **350** may comprise an annual body **351** having inner and outer annual walls **352** and **353** extending between a first open end **354** and an opposing open end **355**. The inner annual wall **352** defines a chamber **356** for housing the diaphragm **340** and a longitudinal axis **357**. The annual body **351** further includes a series of elongated slots **358** circumferentially disposed about the inner annual wall **352** and a series of arcuate-shaped vents **359** formed near the open end **354**.

(60) The elongated slots **358** are configured to receive corresponding stabilizing tabs **329** of the outer hub **320**. The slots **358** are further constructed to dimensions corresponding to the width of the stabilizing tabs to restrict the outer hub **320** from rotating as it is axially translated within the valve housing **350**. The vents **359** are configured to allow air to pass into and/or out of the chamber **356**.

(61) As mentioned above, the outer hub **320** may further include a series of stabilizing tabs **329** formed about the periphery of the annual rim **322** for fitting within a corresponding series of elongated slots formed in the annular wall of the valve housing **350**. The stabilizing tabs **329** cooperate with the elongated slots to guide and assist the axial back-and-forth movement of outer hub **320** along the length of the valve housing **350** and prevent the outer hub **320** from rotating as it is translated within the housing **350**.

(62) A fixed hub **360** having an annular construction is coupled to the inner annual wall **352** in a central portion of the chamber **356**. In the embodiment shown, the fixed hub **360** is integrally formed with the inner annual wall **352**, but in other embodiments, the fixed hub **360** may be coupled to the inner annular wall **352** by bonding, welding, or any other suitable means.

(63) The fixed hub **360** includes an annual rim **361** coupled to a central hub portion **362** by a series of spokes **363** extending therebetween. Each spoke **363** may be substantially triangular in shape,

such that its thickness gradually increases as the spokes **363** extends from central hub portion **362** towards the annual rim **361**. Each spoke **363** may include a spacer guide hole **364** formed approximately midway along its length for receiving a diaphragm spacer of the inner hub **385**, as described in further detail below.

(64) The central hub portion **362** includes a bore **366** extending therethrough. The bore **366** is positioned along the longitudinal axis **357**. The bore **366** is constructed to receive an integrated screw member **370** extending therethrough along the longitudinal axis **357**. As better shown in FIG. **4**, the integrated screw member **370** is an elongated cylindrical body **402** having a center portion **404**, a first end **406**, a first threaded portion **408** extending between the center portion **404** and the first end **406**, an opposing end **410**, and a second threaded portion **412** extending between the center portion **404** and opposing end **410**. In preferred embodiments, the diametrical dimensions of screw member **370** may complement the dimensions of bore **366** such that the center portion **404** of the integrated screw member **370** freely rotates inside of bore **366** as screw member **370** is rotated clockwise or counterclockwise.

(65) As discussed in further detail below, the first threaded portion **408** is threadedly coupled to the outer hub **320** to translate the hub axially along the longitudinal axis **357** as the integrated screw member **370** is rotated clockwise or counterclockwise. Similarly, the second threaded portion **412** is threadedly coupled to inner hub **380** to translate the hub axially along the longitudinal axis **357** as the integrated screw member **370** is rotated clockwise or counterclockwise.

(66) Returning to FIG. **3**, the inner hub **380** comprises an annular construction having an annual rim **381** coupled to a central hub portion **382** by a series of spokes **383** extending therebetween. The central hub portion **382** includes a threaded screw hole **384** for engaging screw member **370** extending therethrough, as described in further detail below. Each spoke **383** may be substantially triangular in shape, such that its thickness gradually increases as the spoke **383** extends from the central hub portion **382** towards the annual rim **381**.

(67) The inner hub **380** may further include a series of cylindrical-shaped diaphragm spacers **385** outwardly extending from a corresponding spoke **383**, and a series of stabilizing tabs **386** formed about the periphery of the annual rim **381**. The diaphragm spacers **385** are constructed to diametrical dimensions that correspond to the diametrical dimensions of the spacer guide holes **364** formed in the spokes **363** of the fixed hub **360**. The diaphragm spacers **385** are constructed to a height that is greater than the thickness of the fixed hub **360**. The diaphragm spacers **385** are, further, configured to mate with and extend through the spacer guide holes **364** as the inner hub **380** is axially translated within the valve housing **350** towards the fixed hub **360**. The spacer **385** aids to align the relative positions of the spokes of the inner hub **380**, fixed hub **360**, and outer hub **320** with one another to limit turbulence and promote laminar airflow through the valve chamber **356**.

(68) The stabilizing tabs **386** are constructed to the dimensions and configured to fit within a corresponding series of elongated slots (not shown) formed in the annular wall of the valve housing **350** to prevent the inner hub **380** from rotating as it is axially translated within the valve housing **350**.

(69) The inner hub **380** may be constructed to have a relatively thin thickness. For example, inner hub **380** may be constructed to a thickness of approximately 0.125 inches. The inner hub **380** may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(70) As better shown in FIG. **4**, the integrated screw member **370** is configured to extend the entire length of the respiratory valve **108** (i.e., from the valve cap indicator **310** to the inner hub **380**) along the longitudinal axis **357**. As shown, at one end, the first end **406** of the screw member **370** may be attached to the adjustable dial **302** at a seat **420** formed in the bottom of the knob **304**. The adjustable dial **302** may be attached to the first end **406** by a sealant, glue, bond, snap-fit, interference fit, or other suitable means. The adjustable dial **302** is configured to rotate the integrated screw member **370** as the knob **304** is manually rotated by the wearer.

(71) On the opposite end of the integrated screw member **370**, opposing end **410** is threadedly coupled to the inner hub **380** at threaded screw hole **384**.

(72) In addition to being attached to the adjustable dial **302** and coupled to the inner hub **380**, the integrated screw member **370** is, further, threadedly coupled to the outer hub **320**. In this way, the first threaded portion **408** of the screw member **370** comprises male threads that correspond to female threads comprising the threaded screw hole **328** of the outer hub **320**. Similarly, the second threaded portion **412** of the screw member **370** comprises male threads that correspond to female threads comprising the threaded screw hole **384** of the inner hub **380**.

(73) As such, when the screw member **370** is rotated by the adjustable dial **302**, the male threads of the first threaded portion **408** engage the female threads of threaded screw hole **328** to translate the outer hub **320** axially along the longitudinal axis **357**. Simultaneously, the male threads of the second threaded portion **412** engage the female threads of threaded screw hole **384** to translate the inner hub **380** axially along the longitudinal axis **357**. Thus, as the screw member **370** is rotated by the adjustable dial **302**, the outer hub **320** and the inner hub **380** are translated axially along the longitudinal axis **357** in tandem, spaced-apart relation.

(74) In some embodiments, the respiratory valve **108** is operable in three modes of operation: an open mode, a vented mode, and closed mode.

(75) FIGS. **4A** and **4B** illustrate the respiratory valve **108** in the open mode. In this mode, the adjustable dial **302** is turned counterclockwise and the inner hub **380** is translated along longitudinal axis **357** such that its annular rim **381** is positioned proximate to or abutting the annular rim **361** of the fixed hub **360**. In this position, the diaphragm spacers **385** of the inner hub **380** extend through the spacer guide holes **364** to lift or urge the outer periphery of the diaphragm **340** away from the fixed hub **360** such that, when the wearer either inhales or exhales, air is permitted to freely and rapidly pass around the diaphragm **340**, as shown by arrows **430**. This permits the wearer to breathe through the valve **108** with minimal effort, thus, improving wearer comfort. This mode, however, provides minimum filtration through the filtering layer **208** of the facemask **102** and is for the breathing comfort of the wearer, only.

(76) As best shown on FIG. **4C**, the lengthwise dimensions of the elongated slots **358** define the distance that the inner hub **380** and the outer hub **320** are permitted to translate axially along the longitudinal axis **357**.

(77) FIGS. **5A** and **5B** illustrate the respiratory valve **108** in the vented mode. In this mode, the adjustable dial **302** is rotated clockwise from the open mode position. As the adjustable dial **302** is rotated clockwise, the inner hub **380** and the outer hub **320** are axially translated downstream in tandem, such that the inner hub **380** is translated axially along longitudinal axis **357** away from the fixed hub **360**, while the outer hub **320** is translated axially along longitudinal axis **357** towards the fixed hub **360**. In this position, the diaphragm spacers **385** of the inner hub **380** are disposed within the spacer guide holes **364** and the diaphragm **340** is no longer being lifted away from the fixed hub **360**. The outer periphery of the diaphragm **340** may be lifted or urged away from the fixed hub **360** by air pressure when the wearer exhales, but drawn towards the fixed hub **360** by negative pressure to form a seal with the annular rim **361** when the wearer inhales. In this mode, exhaled air is permitted to pass around the diaphragm **340** and is exhausted through vents **359**, as shown by arrows **530**, thereby improving user comfort when the wearer exhales. But ambient air is prevented from passing through the valve **108** when the wearer inhales. This permits the wearer to exhale air through the valve **108** with minimal effort, but ambient air must be filtered through the filtering layer **208** of the facemask **102** when the wearer inhales.

(78) FIGS. **6A** and **6B** illustrate the respiratory valve **108** in the closed mode. In this mode, the adjustable dial **302** is rotated clockwise from the vented mode position. As the adjustable dial **302** is further rotated clockwise, the inner hub **380** and the outer hub **320** are axially translated downstream in tandem, such that the inner hub **380** is translated axially along longitudinal axis **357** further away from the fixed hub **360**, while the outer hub **320** is translated axially along

longitudinal axis **357** towards the fixed hub **360** until annular rim **322** is positioned proximate to or abutting the annular rim **361** of the fixed hub **360**. In this position, the diaphragm spacers **385** of the inner hub **380** are disposed within the spacer guide holes **364** and the outer periphery of the diaphragm **340** is sandwiched between annular rim **322** and annular rim **361** such that the diaphragm **340** is secured against the spokes **363** and annular rim **361** of the fixed hub **360**. In this mode, air is prevented from passing in or out of the valve **108** when the wearer inhales and exhales. Thus, air will only pass through the filtering layer **208** of the facemask **102** when the wearer inhales and exhales, for optimum safety.

(79) In sum, to close the valve **108**, the adjustable dial **302** is turned clockwise and the outer hub **320** is advanced towards the center of the valve. When the adjustable dial **302** is completely turned and reaches a stop, the outer hub **320** will have secured the diaphragm **340** against the fixed hub **360**, completely sealing the valve **108**. When the adjustable dial **302** is turned in the opposite direction (i.e., counterclockwise), the outer hub **320** is moved away from the fixed hub **360**, and the inner hub **380** is moved toward the fixed hub **360**. When the adjustable dial **302** is completely turned in the opposite direction and reaches a stop, the diaphragm spacers **385** on the inner hub **380**, lift or urge the diaphragm **340** away from center hub spokes **363** to open the valve **108** for air to flow freely therethrough. In instances when the diaphragm **340** is not secured to the fixed hub **360** by the outer hub **320** and not elevated by the spacers **385** of the inner hub **380**, the valve **108** is vented and allows air to pass through the valve **108** during exhalation, but not during inhalation.

(80) As illustrated in FIGS. **4** through **6**, when the adjustable dial **302** is rotated clockwise from the open mode position to the closed mode position, the distance between the outer hub **320** and the fixed hub **360**, depicted as arrow **460**, gradually narrows until the outer hub **320** abuts the fixed hub **360**. As the distance **460** narrows, the outer periphery of the diaphragm **340** becomes more restricted from being lifted or urged away from the spokes **363** of the fixed hub **360** by air pressure. Thus, as the adjustable dial **302** is rotated clockwise, breathing through the valve **108** becomes more restricted for the wearer until air is no longer permitted to pass in and out of the valve **108** in the closed mode. Therefore, the wearer may adjust the adjustable dial **302** clockwise or counterclockwise until the suitable level of breathing comfort is obtained.

(81) FIG. **7** is a perspective view of another example of a filtering facepiece respirator **700** according to the teaching of the present disclosure. As shown, the facepiece respirator **700** may include a facemask **702**, an upper harness **704** coupled to an upper portion of the face mask **702**, a lower harness **706** coupled to a lower portion of the facemask **702**, and a respiratory valve **708** coupled to a central front portion of the face mask **702**. In preferred implementations, the face mask **702** may include an outer contour **710** defined about the peripheral edge of the facemask **702** to substantially cover the nose and mouth of a wearer and a bendable reinforcement nosepiece **712** coupled to an upper portion of the outer contour **710**. For purposes of simplicity, the facemask **702**, upper harness **704**, and lower harness **706** may be constructed similar to the facemask **102**, upper harness **104**, and lower harness **106** of facepiece respirator **100**, thus, the details of these components will not be described further.

(82) FIG. **8** is an exploded view of respiratory valve **708**. In this example, the respiratory valve **708** is a motorized adjustable bi-mode valve. Moving from right to left, the respiratory valve **708** includes a rotatable pointer **802**, a valve cap indicator **810**, an outer hub **820**, a multifunctional motor module **830**, a diaphragm **840**, a valve housing **850**, and a multifunctional battery module **880**.

(83) As shown, the pointer **802** may comprise an arrow-shaped pointer body **804** and a pointer hub **806**. The pointer **802** be constructed to suitable dimensions complementing indicia inscribed on valve cap indicator **810** to indicate to the wearer what operational mode the respiratory valve **708** is operating in. The pointer **802** may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(84) The valve cap indicator **810** may comprise thin-disc-shaped plate **812** having indicia **814**

inscribed on its upper surface **816**. In the embodiment shown, the indicia **812** includes the wording “VENTED” and “CLOSED” to correspond to the two modes of operation of the respiratory valve **708**. In particular, the indicia **812**, when the pointer **802** is positioned over the indicia, indicates to the wearer the mode of operation that the valve **708** is currently operating in.

(85) The valve cap indicator **810** further includes a screw hole **818** for passing a screw member therethrough, as further described below, and an outer periphery **819**. The outer periphery **819** is preferably constructed to diametrical dimensions that correspond to the diametrical dimensions of a first open end of the valve housing **850** such that the outer periphery **819** mates with an inner wall of the valve housing **850** to enclose the first end.

(86) The valve cap indicator **810** may be constructed from plastic, aluminum, stainless steel, or any other suitable material. The indicia **814** may be inscribed or etched into the upper surface **816** of the valve cap indicator **810** or, in other embodiments, the indicia **814** may include stickers, light-emitting diodes, or may be painted on the of the upper surface **816** of the valve cap indicator **810**.

(87) Moving further downstream, the outer hub **820** comprises an annular construction having an annual rim **822** coupled to a central hub portion **824** by a series of spokes **826** extending therebetween. The central hub portion **824** includes a threaded screw hole **828** for engaging a screw member extending therethrough. Each spoke **826** may be substantially triangular in shape, such that its thickness gradually increases as the spoke **826** extends from the central hub portion **824** towards the annual rim **822**.

(88) The outer hub **820** may further include a series of stabilizing tabs **829** formed about the periphery of the annual rim **822** for mating with a corresponding series of elongated slots **858** formed in the annular wall of the valve housing **850**. The stabilizing tabs **829** cooperate with the elongated slots **858** to guide the axial back-and-forth movement of the outer hub **820** along the length of the valve housing **850** and prevent the outer hub **820** from rotating as it is translated inside of the housing **850**.

(89) The outer hub **820** may be constructed to have a relatively thin thickness. For example, outer hub **820** may be constructed to a thickness of approximately 0.125 inches. The outer hub **820** may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(90) The multifunctional motor module **830** may include a thin disc-shaped motor **832**, a thin disc-shaped motor base **834**, and an orifice **836** extending therethrough. The motor module **830** serves to secure an inner peripheral region of the diaphragm **840** to a central support structure in the valve housing **850** and to rotate an integrated screw member extending through the center of the valve **708**, as described in further detail below. The motor **832** may comprise, for example, a low-profile rotary solenoid actuator, and the motor base **834** may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(91) Next, the diaphragm **840** may include a thin disc-shaped body **842** having an inner peripheral region **844**, an outer peripheral region **846**, and an orifice **848** extending therethrough. Similar to diaphragm **340**, diaphragm **840**, when secured in close proximity to a central hub housed within the valve housing **850**, forms a hermetic seal to prevent airflow through the respiratory valve **708**, as described in further detail below. The diaphragm **840** is preferably constructed from rubber, latex, polymers, or any other non-porous pliable material. In other embodiments, the diaphragm **840** may be constructed from filtering media.

(92) Moving further downstream, the valve housing **850** may comprise an annual body **851** having inner and outer annual walls **852** and **853** extending between a first open end **854** and an opposing open end **855**. The inner annual wall **852** defines a chamber **856** for housing the diaphragm **840**, motor module **830**, and outer hub **820**, and a longitudinal axis **857**. The annual body **851** further includes a series of elongated slots **858** circumferentially disposed about the inner annual wall **852** and a series of arcuate-shaped vents **859** formed near the open end **854**.

(93) The elongated slots **858** are configured to receive corresponding stabilizing tabs **829** of the outer hub **820**. The slots **858** are further constructed to dimensions corresponding to the width of

the stabilizing tabs **829** to restrict the outer hub **820** from rotating as it is axially translated within the valve housing **850**. The vents **859** are configured to allow air to pass into and/or out of the chamber **856**.

(94) A fixed hub **860** having an annular construction is coupled to the inner annular wall **852** in an aft portion of the chamber **856**. In the embodiment shown, the fixed hub **860** integrally formed with the inner annular wall **852**, but in other embodiments, the fixed hub **860** may be coupled to the inner annular wall **852** by bonding, welding, or any other suitable means.

(95) The fixed hub **860** includes an annular rim **862** coupled to a central hub portion **864** by a series of spokes **866** extending therebetween. Each spoke **866** may be substantially triangular in shape, such that its thickness gradually increases as the spokes **866** extends from central hub portion **864** towards the annular rim **862**.

(96) The central hub portion **864** includes a tap hole **868** extending substantially. The tap hole **868** is positioned along the longitudinal axis **857**. The tap hole **868** is constructed to receive an integrated screw member **870** extending therethrough along the longitudinal axis **857**. As better shown in FIG. **9B**, the integrated screw member **870** is an elongated cylindrical body **902** having a first end **904**, an opposing end **906**, a threaded portion **908** extending between the first end **904** and the opposing end **906**.

(97) Returning to FIG. **8**, the multifunctional battery module **880** may comprise a body **882** comprising a thin-disc construction. In other embodiments, the body **882** of the multifunctional battery module **880** may be constructed to other geometrical shapes. As better shown in FIG. **9B**, the multifunctional battery **880** may be coupled to a surface of the central hub portion **864** by bonding, welding or other mechanical means. The multifunctional battery module **880** may be electrically coupled by wiring to the multifunctional motor module **830** to power the motor **832** and electronic circuitry (not shown) controlling the motor **832**. In other embodiments, the multifunctional battery module **880** and the multifunctional motor module **830** may include other features, such as, but not limited to, wireless communications between the facepiece and a mobile device or computer and the monitoring of air quality within the facemask.

(98) The automated respiratory valve **708** may be operable in two modes of operation: a vented mode and closed mode. FIGS. **9A** and **9B** illustrate the respiratory valve **708** in the vented mode.

(99) As shown, the integrated screw member **870** is configured to extend the entire length of the respiratory valve **708** (i.e., from the valve cap indicator **810** to the fixed hub **860**) along the longitudinal axis **857**. At one end, the opposing end **906** of the screw member **870** may be attached to the rotatable pointer **802** at a seat **920** formed in the bottom of the pointer hub **806**. The pointer **802** may be attached to opposing end **906** by a sealant, glue, bond, snap-fit, interference fit, or other suitable means. The motor **832** is configured to rotate the integrated screw member **870** and the attached pointer **802** between a vented mode position and a closed mode position via electronic circuitry (not shown).

(100) On the opposite end, first end **904** may be received by the tap hole **868** in the central hub portion **864** of the fixed hub **860**. In some embodiments, the tap hole **868** may be dimensioned such that the first end **904** of the integrated screw member **870** is permitted to rotate freely within hole **868**. In other embodiments, the first end **904** of the integrated screw member **870** may be rotatably coupled to roller bearings installed in the tap hole **868**.

(101) As further shown, the inner peripheral region **844** of the diaphragm **840** may be fixedly attached to a surface of the central hub portion **864** by, for example, glue, sealant, or bonding. Similarly, the base **834** of the motor module may be fixed atop the inner peripheral region **844** of the diaphragm **840** by, for example, glue, sealant, or bonding. According to this construction, the inner peripheral region **844** of the diaphragm **840** is fixed in place, while the outer peripheral region **846** of the diaphragm **840** may be permitted to be urged or otherwise moved away from spokes **866** and annular rim **862** of the fixed hub **860**, thus permitting air to flow around the diaphragm **840** and through the valve chamber **856**.

(102) Upstream, the male threads of the threaded portion **908** of the screw member **870** are configured to engage the female threads of the threaded screw hole **828** of the outer hub **820** to translate the hub axially along the longitudinal axis **857** as the screw member **870** is rotated clockwise or counterclockwise. The stabilizing tabs **829** (FIG. **8**) are configured to mate or fit within the elongated slots **858** to guide the outer hub **820** as it translates within the chamber **856** and, further, prevents the outer hub **820** from rotating about the screw member **870** as the outer hub **820** is axially translated.

(103) In the vented mode, the outer hub **820** is positioned along longitudinal axis **857** such that its annual rim **822** of the outer hub **820** is spaced apart from the annular rim **862** of the fixed hub **860**. In this position, the spacing between annual rim **822** and annular rim **862** permits the outer peripheral region **846** of the diaphragm **840** to be lifted or urged away from the annular rim **862** and spokes **866** of the fixed hub **860** by air pressure when the wearer exhales, however, the outer peripheral region **846** of the diaphragm **840** is drawn towards the fixed hub **860** by negative pressure to form a seal with annual rim **862** when the wearer inhales. In this mode, exhaled air is permitted to pass around the diaphragm **840** and is exhausted through vents **859**, as shown by arrows **930**, thereby improving user comfort when the wearer exhales. But ambient air is prevented from passing through the valve **708** when the wearer inhales. This permits the wearer to exhale air through the valve **708** with minimal effort, but ambient air must be filtered through the filtering layer of the facemask **702** when the wearer inhales.

(104) FIGS. **10A** and **10B** illustrate respiratory valve **708** in the closed mode. In this mode, the integrated screw member **870** and the attached pointer **802** are rotated clockwise from the vented mode position by a controller (not shown) electrically connected to the motor module **830**. In particular, as the controller is commanded by the wearer to operate in the closed mode, the controller transmits an electric signal to the motor module **830** to start the motor **832** and the motor **832** rotates the screw member **870** clockwise from a vented position to a closed position. Commands may be transmitted by the wearer to the controller by, for example, wireless connection, such as a mobile application stored on and activated from a mobile device. When the wearer decides that it wants to return the mask back to its vented mode of operation, the wearer may activate the controller to reverse the motor **832**, thus rotating the screw member **870** counterclockwise from the closed position to the vented position.

(105) Other embodiments may include additional functions of the multifunctional motorized motor module **830** include but are not limited to communication to and from the mask and sensing of conditions within the mask.

(106) As the screw member **870** is rotated clockwise, the male threads of the threaded portion **908** of the screw member **870** engage the female threads of the threaded screw hole **828** of the central hub portion **824** of the outer hub **820** to axially translate the outer hub **820** towards the fixed hub **860**. When the outer hub **820** is translated to the closed mode position, the annual rim **822** of the outer hub **820** is positioned proximate to or abutting the annular rim **862** of the fixed hub **860**. In this position, the outer peripheral region **846** of the diaphragm **840** is sandwiched between the annular rim of the outer hub **822** and the annular rim of the fixed hub **862** such that the diaphragm **840** is secured against the annual rim of the outer hub **822** and annular rim of the fixed hub **862** and the spokes **866** of the fixed hub **860**. In this mode, the valve **708** is completely sealed and air is prevented from passing in or out of the valve **708** when the wearer inhales and exhales. Thus, air will only pass through the filtering layers of the facemask **702** when the wearer inhales and exhales, for optimum safety.

(107) FIG. **11** is a perspective view of a third example of a filtering facepiece respirator **1100** according to the teaching of the present disclosure. As shown, the facepiece respirator **1100** may include a facemask **1102**, an upper harness **1104** coupled to an upper portion of the face mask **1102**, a lower harness **1106** coupled to a lower portion of the facemask **1102**, and a respiratory valve **1108** coupled to a central front portion of the face mask **1102**. In preferred implementations, the face

mask **1102** may include an outer contour **1110** defined about the peripheral edge of the facemask **1102** to substantially cover the nose and mouth of a wearer and a bendable reinforcement nosepiece **1112** coupled to an upper portion of the outer contour **1110**. For purposes of simplicity, the facemask **1102**, upper harness **1104**, and lower harness **1106** may be constructed similar to the facemask **102**, upper harness **104**, and lower harness **106** of facepiece respirator **100**, thus, the details of these components will not be described further.

(108) As shown in this example, the respiratory valve **1108** is a bi-mode valve with a rotatable valve gate that's operable in a vented mode of operation and a closed mode of operation. As shown, respiratory valve **1108** may include a valve housing **1120**, a rotatable valve gate **1130** having an arm portion that's slidable within a slot **1122** formed along the periphery of the valve housing **1120**, and a valve cap indicator **1126**.

(109) In the example shown, the valve housing **1120** comprises a thin, disc-shaped construction, but in other embodiments, the construction of the valve housing **1120** may include other suitable geometric shapes. The valve housing **1120** further includes one or more vents **1124** for expelling air out of valve **1120**. The valve housing **1120** and its components may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(110) The valve cap indicator **1126** may comprise a thin-disc-shaped plate having indicia inscribed on its upper surface. In the embodiment shown, the indicia includes the wording "VENTED" and "CLOSED" to correspond to the two modes of operation of the respiratory valve **1108**. The valve cap indicator **1126** may include an outer periphery **1128** preferably constructed to diametrical dimensions that correspond to the diametrical dimensions of valve housing **1120** such that the outer periphery **1128** mates with an open end of the valve housing **1120** to enclose housing.

(111) The valve cap indicator **1126** may be constructed from plastic, aluminum, stainless steel, or any other suitable material. The indicia may be inscribed or etched into the upper surface of the valve cap indicator **1126**. In other embodiments, the indicia may include stickers, light-emitting diodes, or may be painted on the upper surface of the valve cap indicator **1126**.

(112) As mentioned above, the manual bi-mode respiratory valve **1108** may be operable in two modes of operation: a vented mode and closed mode. FIGS. **12A** and **12B** illustrate the respiratory valve **1108** in the vented mode.

(113) As shown, the valve gate **1130** may include a gate body **1202**, an integrated gate arm **1204**, and an arm cap **1206**. In the example shown, the gate body **1202** comprises a substantially semi-circular-shaped construction. The gate body **1202** is rotatably coupled to the valve housing **1120** by a pivot **1210**. The pivot **1210** may include a dowel pin, bearing assembly, or other suitable means.

(114) Respiratory valve **1108** further includes a diaphragm **1140**. As shown, the diaphragm **1140** comprises a substantially elongated body **1142** having a fixed end **1302** (FIG. **13**) and a free end **1144**. The diaphragm **1140** is preferably constructed from rubber, latex, polymers, or any other non-porous pliable material.

(115) In other embodiments, the diaphragm **1140** may be constructed from filtering media. In particular, the diaphragm **1140** may be made of porous material that filters microscopic particulates. In such embodiments, air is being filtered through the facemask **1102** and the respiratory valve **1108**.

(116) The fixed end **1302** may be secured to the valve housing **1120** by glue, bonding, or other suitable means. The free end **1144**, shown in this example as being semi-circular in shape, is configured such that it may be urged or otherwise moved away from the valve housing **1120** under positive pressure to permit air to pass from the valve **1108** when the wearer exhales. But the free end **1144** is also configured such that it may be drawn toward the valve housing **1120** under negative pressure to create a seal between the free end **1144** and the valve housing **1120** that prevents air from passing into the valve **1108** when the wearer inhales.

(117) Respiratory valve **1108** further includes a set of arcuate-shaped stops **1150**. The stops are configured to restrict the rotational movement of the valve gate **1130** about the pivot **1210**.

(118) In the vented mode, as shown in FIG. 12B, the gate body **1202** is rotated clockwise about the pivot **1210** by the arm cap **1206** such that the gate body **1202** is positioned over the fixed end **1302** (FIG. 13B) of the diaphragm **1140** while the free end **1144** is uncovered and permitted to be urged away from the valve housing **1120** under positive pressure. This allows air to flow freely through the valve **1108** when the wearer exhales, as shown by arrows **1160**, but air is prevented from flowing through the valve **1108** when the wearer inhales.

(119) FIGS. 13A and 13B illustrate the respiratory valve **1108** in the closed mode. In the closed mode, the gate body **1202** is rotated counterclockwise about the pivot **1210** by engaging the arm cap **1206** such that the gate body **1202** is positioned over the free end **1144** of the diaphragm **1140**, thus abutting the free end **1144** against the valve housing **1120**. This prevents air from flowing through valve **1108** when the wearer inhales or exhales. Therefore, air will only pass through the filtering layers of the facemask **1102** when the wearer inhales and exhales, for optimum safety.

(120) FIG. 14 is a perspective view of a fourth example of a filtering facepiece respirator **1400** according to the teaching of the present disclosure. As shown, the facepiece respirator **1400** may include a facemask **1402**, an upper harness **1404** coupled to an upper portion of the face mask **1402**, a lower harness **1406** coupled to a lower portion of the facemask **1402**, and a respiratory valve **1408** coupled to a central front portion of the face mask **1402**. In preferred implementations, the face mask **1402** may include an outer contour **1410** defined about the peripheral edge of the facemask **1402** to substantially cover the nose and mouth of a wearer and a bendable reinforcement nosepiece **1412** coupled to an upper portion of the outer contour **1410**. For purposes of simplicity, the facemask **1402**, upper harness **1404**, and lower harness **1406** may be constructed similar to the facemask **102**, upper harness **104**, and lower harness **106** of facepiece respirator **100**, thus, the details of these components will not be described further.

(121) FIG. 15 is an exploded view of respiratory valve **1408**. In this example, the respiratory valve **1408** is a bi-mode valve with a slidable valve gate that's operable in a vented mode of operation and a closed mode of operation. As shown, respiratory valve **1408** may include a valve housing **1500**, a diaphragm **1520**, a slidable valve gate **1530**, and a valve cap indicator **1540**.

(122) In the example shown, the valve housing **1500** comprises an annular wall **1502** and an internal landing region **1504**. The annular wall **1502** and the landing region **1504** define a cell **1506** that houses the diaphragm **1520** and valve gate **1530**. The valve housing **1500** further includes a pair of slots **1508** and vents **1510** formed in the annular wall **1502**, a substantially semi-circular orifice **1512** extending through the center of the landing region **1504**, and series of arcuate-shaped stops **1514** formed along the inner surface of the annular wall **1502**. The valve housing **1500** is shown comprising disc-shaped construction, but in other embodiments, the construction of the valve housing **1500** may include other suitable geometric shapes. The valve housing **1500** and its components may be constructed from plastic, aluminum, stainless steel, or any other suitable material.

(123) The diaphragm **1520** may comprise an extended arch-shaped body **1522** having a fixed end **1524** and a free end **1526**. The diaphragm **1520** is preferably constructed from rubber, latex, polymers, or any other non-porous pliable material. In other embodiments, the diaphragm **1520** may be constructed from filtering media

(124) As shown, the diaphragm **1520** is configured to be assembled within an arch-shaped recess **1516** formed in the landing region **1504** of the valve housing **1500**. When assembled within the recess **1516**, the fixed portion **1524** of the diaphragm **1520** may be affixed to a seat portion **1517** of the recess **1516**, while the free end **1526** may be configured to rest atop a rim portion **1518** of the recess **1516**. In this way, when unrestricted, the free end **1526** of the diaphragm **1520** may be urged away from the rim portion **1518** of the recess **1516** under positive pressure to permit air to pass from the valve **1408** when the wearer exhales. But the free end **1526** is also configured such that it may be drawn toward the rim portion **1518** under negative pressure to create a seal between the free end **1526** and the rim portion **1518** that prevents air from passing into the valve **1408** when the

wearer inhales.

(125) The valve gate **1530** may include an elongated plate member **1532** having a top arc-shaped tab **1534**, a bottom arc-shaped tab **1536**, and a cut-out **1538** formed in the middle of the plate member **1532**. The arc-shaped ends **1534** may be chamfered to define bearing surfaces **1533** that engage stops **1514**. The valve gate **1530** may be assembled over the diaphragm **1520** to retain the diaphragm in the recess **1516** of the landing region **1504**.

(126) The valve cap indicator **1540** may comprise a thin-disc-shaped plate **1542** having indicia **1544** inscribed on its upper surface **1546**. In the example shown, the indicia **1544** includes the wording “VENTED” and “CLOSED” to correspond to the two modes of operation of the respiratory valve **1408**. The valve cap indicator **1540** may further include an outer periphery **1548** preferably constructed to diametrical dimensions that correspond to the diametrical dimensions of the inner diameter of the annular wall **1502** such that the outer periphery **1548** mates with an annular wall **1502** to enclose the cell **1506** of the valve housing **1500**.

(127) The valve cap indicator **1540** may be constructed from plastic, aluminum, stainless steel, or any other suitable material. The indicia **1544** may be inscribed or etched into the upper surface **1546** of the valve cap indicator **1540**. In other embodiments, the indicia **1544** may include stickers, light-emitting diodes, or may be painted on the upper surface **1546** of the valve cap indicator **1540**.

(128) Respiratory valve **1408** may be operable in two modes of operation: a vented mode and closed mode. FIGS. **16A** and **16B** illustrate the respiratory valve **1408** in the vented mode.

(129) In the vented mode, the top tab **1534** of the valve gate **1530** is engaged by the wearer to push the plate member **1532** downward such that the cut-out **1538** is positioned over the free end **1526** of the diaphragm **1520**. The vertical translation of the valve gate **1530** is restricted by stops **1514**. In this position, the free end **1526** of the diaphragm **1520** is unrestricted and may be urged away from the rim portion **1518** (FIG. **15**) of the recess **1516** under positive pressure to permit air to pass from the valve **1408** when the wearer exhales, as shown by arrows **1602**. Alternatively, the free end **1526** may be urged or drawn toward the rim portion **1518** (FIG. **15**) under negative pressure to create a seal between the free end **1526** and the rim portion **1518** that prevents air from passing into the valve **1408** when the wearer inhales.

(130) FIGS. **17A** and **17B** illustrate respiratory valve **1408** in the closed mode. In the closed mode, a bottom tab **1536** of the valve gate **1530** is engaged by the wearer to push the plate member **1532** upwards such that the plate member **1532** is positioned over the free end **1526** of the diaphragm **1520**, thus abutting the free end **1526** against the rim portion **1518** (FIG. **15**) of the recess **1516**. The upward vertical translation of the valve gate **1530** is restricted by stops **1514**. Because the free end **1526** is secured against the rim portion **1518** (FIG. **15**), a seal is created between the free end **1526** and the rim portion **1518** (FIG. **15**) **1120** air is prevented from flowing through the valve **1408** when the wearer inhales or exhales. Therefore, air will only pass through the filtering layers of the facemask **1402** when the wearer inhales and exhales, for optimum safety.

(131) The present invention may be implemented in various embodiments, namely, manual, automatic, or a combination of manual and automatic. While the present invention depicts a screw mechanism to open and close the valve, there are many other potential methods that could be utilized. For example, instead of a screw, the hubs may be advanced and retracted along a rod utilizing a push or pull mechanism. In other embodiments, the hubs may be translated by other sliding or rotating mechanisms.

(132) In some embodiments, the filtering facepiece respirator may also include smart bio-sensors that detect and measure intra-mask oxygen saturation levels, carbon dioxide levels, nucleic acid levels, the volume of expired or inspired air, as well as detect air leakage from the facemask. In these embodiments, the smart sensors may report data to a central processing unit, where the central processing unit uses the smart sensor data to alert the user of a change in mask environment conditions.

(133) Other embodiments may be fitted with RF technology that provides communication between

the respirator and a mobile device or central processing unit to control the respirator's mode of operation. In these embodiments, the controller may be integrated for use within public spaces such as offices, hospitals, clinics, restaurants, public and private buildings, service businesses, gyms and health clubs, and retailers.

(134) In some embodiments, the entire respirator may be discarded. In other embodiments, the filtering layer material may be replaceable. In other embodiments, the respiratory valve may be replaceable while the facemask is preserved.

(135) While the embodiments described in the present disclosure teach respiratory valves generally having annular-shaped construction, other respiratory valves and components according to the teachings of the present disclosure may include square-spaced, triangular-shaped, pentagon-shaped, or other geometrical-shaped constructions.

(136) In general, terms such as “coupled to,” and “configured for coupling to,” and “secured to,” and “configured for securing to” and “in communication with” (for example, a first component is “coupled to” or “is configured for coupling to” or is “configured for securing to” or is “in communication with” a second component) are used herein to indicate a structural, functional, mechanical, electrical, signal, optical, magnetic, electromagnetic, ionic or fluidic relationship between two or more components or elements. As such, the fact that one component is said to be in communication with a second component is not intended to exclude the possibility that additional components may be present between, and/or operatively associated or engaged with, the first and second components.

(137) Although the previous description illustrates particular examples of various implementations, the present disclosure is not limited to the foregoing illustrative examples. A person skilled in the art is aware that the disclosure as defined by the appended claims and their equivalents can be applied in various further implementations and modifications. In particular, a combination of the various features of the described implementations is possible, as far as these features are not in contradiction with each other. Accordingly, the foregoing description of implementations has been presented for purposes of illustration and description. Modifications and variations are possible in light of the above description.

Claims

1. A filtering facepiece respirator comprising: a facemask adapted to fit over the nose and mouth of a wearer, the facemask comprising a mask body containing a filtering structure; a harness coupled to the mask body for securing the facemask on the face of the wearer; and a mechanical valve coupled to a portion of the mask body proximate the wearer's mouth, the valve being manually adjustable between a first mode of operation and a second mode of operation, where in the first mode of operation air is permitted to flow through the valve on exhalation but not on inhalation and in the second mode of operation air is obstructed from flowing through the valve on exhalation and inhalation, where the valve comprises a valve housing having an interior chamber, a fixed hub member disposed within the chamber, a first movable hub member disposed within the chamber, a pliable diaphragm disposed between the fixed hub member and the first movable hub member, a screw member disposed within the chamber extending along a longitudinal axis of the valve housing, and an adjustable dial, the adjustable dial being rotatable about the longitudinal axis, where the adjustable dial is coupled to the screw member and the screw member is threadedly coupled to the first movable hub member such that when the adjustable dial is rotated by the wearer, the screw member is also rotated and the threaded engagement between the screw member and the first movable hub member causes the first movable hub member to translate axially along the longitudinal axis away from or towards the fixed hub member.

2. The filtering facepiece respirator of claim 1, where the mask body comprises a first porous layer and a filtering layer.

3. The filtering facepiece respirator of claim 1, where the mask body comprises a first porous layer, a second porous layer, and a filtering layer, where the filtering layer is disposed between the first porous layer and the second porous layer.
 4. The filtering facepiece respirator of claim 1, where in the first mode of operation the first movable hub member is spaced apart from the fixed hub member, thus permitting an outer periphery of the diaphragm to be urged away from the fixed hub member under positive pressure and air is allowed to freely flow through the valve when the wearer exhales.
 5. The filtering facepiece respirator of claim 4, where the outer periphery of the diaphragm is drawn towards the fixed hub member under negative pressure to obstruct the flow of air through the valve when the wearer inhales.
 6. The filtering facepiece respirator of claim 1, where in the second mode of operation the first movable hub member abuts the fixed hub member to secure the diaphragm therebetween, thus retaining an outer periphery of the diaphragm in contact with the fixed hub member to create a seal therebetween that obstruct air from flowing through the valve when the wearer exhales or inhales.
 7. The filtering facepiece respirator of claim 1, where the valve may be adjusted to a third mode of operation, where in the third mode of operation air is permitted to flow through the valve when the wearer inhales or exhales.
 8. The filtering facepiece respirator of claim 1, where the valve may be adjusted from one mode of operation to another mode of operation by a switch, slidable gate, rod, rotating arm, lever, button, toggle, dial, knob, joystick, rheostat, or electronic controller.
 9. The filtering facepiece respirator of claim 1 further comprising a valve cap indicator coupled to a front portion of the valve housing, the valve cap indicator comprising a plate of material inscribed with indicia indicating the state of airflow through the valve utilizing a colored indicator, label, light indicator, auditory system, or display.
 10. The filtering facepiece respirator of claim 1, where the diaphragm is made of a material that filters microscopic particulates.
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